Cloning-free long-read profiling of somatic hypermutation

Two biological systems test one control-based Oxford Nanopore workflow

1 Complementary test systems

Mouse GC B cells: J_{H4} downstream intron

~1.23-kb amplicon

- Wt ($n = 3$)
- *Aicda*$^{-/-}$ ($n = 3$)

Same mice benchmarked by cloned-amplicon Sanger and ONT

N114 hybridoma

Fixed immunoglobulin V region

N114

- pMX-*Aicda*
- pMX-Empty vector

~0.65-kb amplicon

$n = 3$ independent subclones per condition

2 Shared ONT workflow

R10.4.1 | Kit 14 | SUP basecalling

Intact amplicons are retained as single-molecule reads



substitutions remain phased within each read

Read QC — length ≤ max EE ≤ 0.01/base

Align — vs. paralogue; chimeras removed

Depth — random 500 reads

Call SNVs — tested against control errors

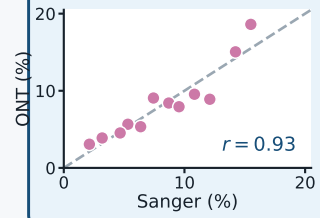
SHMtrace: Benjamini–Hochberg FDR ≤ 0.05; sites at ≥ 50% removed
Each position–substitution counted once per sample

Control-based interpretation

Aicda$^{-/-}$ or empty-vector controls measure residual background

3 Validated outcomes

Sanger concordance



AID dependence

Mouse J_{H4} ≥137×

N114 V region ≥92×

G:C-biased AID signature

Longer region

J_{H4}: 1.23-kb intact reads

529-bp Sanger window

1,141-bp reliable analysis

long-range linkage retained

Depth + background

94% J_{H4} | **91%** N114

variants recovered at 500 reads

Aicda$^{-/-}$ control background

≈0.48 × 10^{-5} variants/bp

Cloning-free • intact long molecules • depth-standardised, substitution-focused profiling